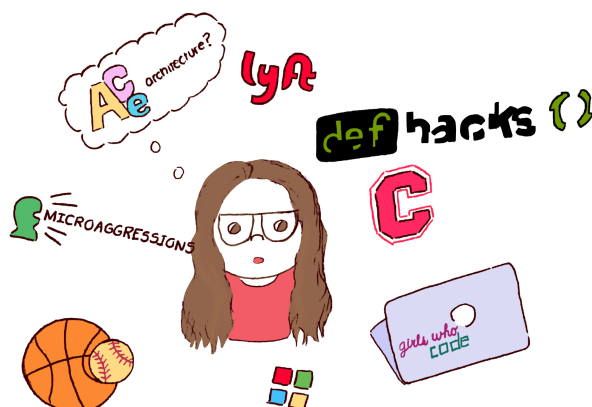


Stuyvesant Alumni Perspectives: Caitlin Stanton, Class of 2016

Isabelle Lam

As a female in STEM, Caitlin Stanton (Class of 2016) shares her viewpoints on several challenges and successes during high school and college.



1 Memories of Stuy

When Caitlin Stanton first started at Stuyvesant, she truly thought she would pursue a career in architecture. To fuel this interest, she took advantage of a national career immersion opportunity named ACE (Architecture-Construction-Engineering) Mentor Program of America. Surprisingly, this program showed her that she didn't really want to study architecture after Stuyvesant. Although she stayed with the program for all four years of high school, Caitlin realized that she enjoyed only certain parts of architecture, such as the artistic and functional aspects.

When her parents enrolled her in a Girls Who Code (GWC) program, Caitlin found a supportive environment to learn about STEM. Although she initially wasn't excited about attending this seven-week program, she discovered that she enjoyed learning in a STEM environment. In her Stuyvesant career, she initially didn't enjoy Introduction to Computer Science, since the processes in Computer Science (CS) are much different from other classes. She recalled, "Debugging my final project [...] was a process that I hadn't been used to before. Most other work you do in high school was kinda just cut-and-dried, like you read a history

textbook or solve a math equation. There's not a lot that you can fix, not a lot of problems you can solve. Code is a whole different story." Instead of just memorizing, regurgitating, and then promptly forgetting what she learned, CS challenged her to really understand her code. It was a "really different experience from the rest of high school," she said.

Despite this newfound interest in CS, Caitlin pursued other extracurriculars while attending Stuyvesant. She was one of the co-captains of the softball team, participated in ARISTA, and was the basketball team manager. She explained, "I knew that I liked volunteering and pushing myself academically, so ARISTA made a lot of sense. I also knew I enjoyed sports like basketball and softball. It was nice to have a chunk of my time put aside for things that I knew I enjoyed, outside of going home and doing homework." She advised against padding college applications by joining a lot of clubs, especially if you don't have any interest in those clubs. She noted how a lot of her peers at Stuy exaggerated their involvement in clubs without contributing much and joined just for the sake of listing 15 clubs on their resume. She remarked, "It's not necessarily worth spending your time doing things that you don't enjoy just to get into a certain college. You don't want to look back on high school and think, 'Wow, I spent my time doing things I didn't like, all for nothing.'"

In her senior year, she co-founded a hackathon named def hacks() with her friend Emily Redler. After observing a "I don't belong here" mentality coupled with being a "16 or 17 year old girl" at traditional hackathons, Caitlin realized that she should create her own. This new hackathon would not be limited to students in specialized high schools and would not instill a sense of inferiority, one of her struggles. She noted, "I wanted to create a space for high school students, by high school students." At first, it was difficult getting sponsorships for venues and spreading the word about this new organization. Caitlin explained, "We had a bunch of people we could reach out to. All we really had to do was prove to them that our concept was worth it." Many times companies didn't want to donate to def hacks() because they were just highschoolers, and they didn't have any other sponsors. But after the first company, ThoughtWorks, made a contribution, other companies' donations began trickling in quite easily. Despite how arduous this experience was, Caitlin found it very rewarding. Both she and Emily were able to gain a new skillset and many new experiences. For example, some companies were hesitant to provide large sums of money, but Caitlin was able to negotiate smaller sums, preventing a net loss for her hackathon. After the first event, companies were more open to sponsoring, since she was able to show substantial participation in her hackathon.

2 Majoring in CS in College, Confronting Bias

In college life at Cornell University, Caitlin has encountered biases against women within her major, Electrical and Computer Engineering (ECE). She explained how she's "lucky enough to never have been outright discriminated against," but, "what's more of the norm is unconscious bias or microaggressions. A lot of my male friends will not turn as quickly to me for homework help, they'll interrupt me while I'm talking, or they'll restate the ideas that I said five minutes ago." She elaborated, stating, "It also shows how maybe that person didn't mean anything by that comment, but it came so easily to them to say." She recognizes that "most of the time, these are my friends. I know that they don't actively mean anything by it, it's just the way that society has raised them. The environment they were put in places women so they feel they can't be a part of the STEM community because they can't have more 'masculine' traits or personalities." She proposes teaching girls from a young age about their self-worth and providing adequate and easily accessible resources and opportunities for girls.

Although Cornell boasts a 50/50 female to male ratio, that ratio only applied to beginner STEM courses.

Caitlin noticed the ratio becoming smaller and smaller as she took more advanced courses. To bridge the gap, some of her professors have taken steps to try to solve this problem. She reminisced, “Last year, I had a professor for a notoriously difficult CS course who said at one of the first lectures, ‘Hey everyone, you belong here. You belong in this class.’ [...] Having someone who I didn’t know, and who is a white, cis-man, tell me, ‘Hey, this is a space for you. I actively want to make this inclusive and I actively want to help everyone think that this is a class that they can take and succeed in.’ That meant a lot.”

Other teachers were uplifting in more quiet ways. Her Fast Robots professor, a female, noticed that there was only one woman enrolled in the course, and emailed all of the women of the ECE Department with details about the course. Despite this thoughtful gesture on the professor’s part, Caitlin lamented that “the fact that I had to have a professor reach out to me to find a certain class and to enroll in it is kinda sad.” Due to the professor’s efforts, the ratio of women to men in the class trended upwards and was 1:3 at the time of interview.

One professor took a non-confrontational approach. Caitlin recalled how a male student in her class who outright refused to ask female teaching assistants (TAs) for help. The professor didn’t correct the male student, but gave TAs permission to kick someone out of office hours for rudeness, which felt empowering. Caitlin also noted how some females in her engineering sorority did not have a single female engineering professor at Cornell. In contrast, Caitlin has had two so far.

As for Cornell’s overall effort to eliminate bias, she commented, “I think Cornell is taking strides. I don’t think they’re big enough yet though. There’s a long way to go for them to be able to say that the college in general is an equitable place for people of any gender and people of any identity.”



3 The Working World and Beyond

Outside of academic life, Caitlin enjoyed largely positive internships at technology titans such as Lyft and Microsoft where she benefited from supportive and instructive environments. In the summer of 2019, she worked as an electrical engineering intern for the first time at Microsoft. In this new environment, she was expected to troubleshoot Bluetooth and WiFi for the Microsoft Surface laptops. At Lyft, where Caitlin interned virtually in June and July 2020, it was her first time combining machine learning (ML) with

hardware. Before this internship, she had thought of ML as a software concept and not implementation into hardware. She was tasked with researching ML chips for the car and looking for potential partner companies. Although she was given a large amount of responsibility in making a chip recommendation, she felt that the people at Lyft were very receptive and willing to help. This would be an ideal environment for younger girls to nurture interests in STEM.

Caitlin is currently pursuing a Masters of Engineering at Cornell, a one-year program that is more focused on a project than on research. She describes ECE as “[doing] a great job of illustrating the wide range of field from lowest level hardware in silicon chip design to highest level software in user-facing apps.” After she earns her master’s, she hopes to go into a Hardware and Engineering role, where she can build circuits and code up hardware. For circuits, she plans to intern at Samsung Research America (S.R.A.) this coming summer, where she will be working with emerging technology as part of the Think Tank Team. Hardware is the branch of CS that focuses on “the process of designing, developing, testing and producing computer systems and various physical components related to computer systems.” She thinks companies such as Apple, Samsung, and Google are always looking for hardware engineers with innovative ideas. However, she doesn’t want to limit herself to technical pursuits. She plans to go into public speaking and continue advocating for STEM because she really enjoys sharing her experiences with larger audiences. Long term, Caitlin plans to explore entrepreneurship in business, marketing, and sustainability as her way to make a mark on the world. To do this, she envisions becoming an executive manager or board member so she can work with groups of people, share ideas, and keep her career both technical and non-technical.

4 Statistics and References

The national average female to male ratio in STEM courses is 17%. The Society of Women Engineers provides more in-depth statistics, reporting that “over 32% of women switch out of STEM degrees in college” with “only 30% of the [68% or less of women] who earn bachelor’s degrees in engineering still working in engineering 20 years later.”

According to Cornell, the amount of women in ECE has increased to 35% in 2017 from 13% in 2012. They also note that “for the first time in Cornell Engineering’s history, the college has admitted a class composed of more women than men, with the Class of 2021 being 51% female.” These statistics fall within the gender binary and do not cover the full gender spectrum as accepted today.

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